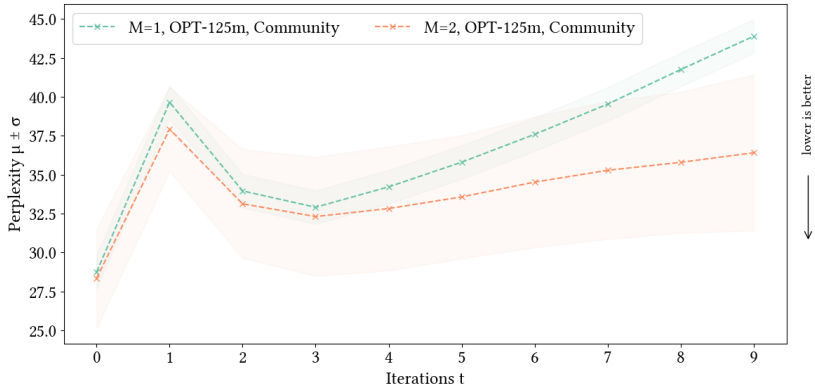
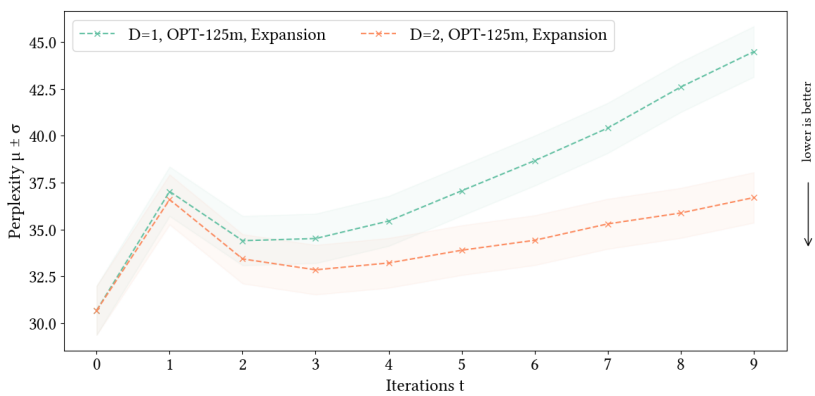
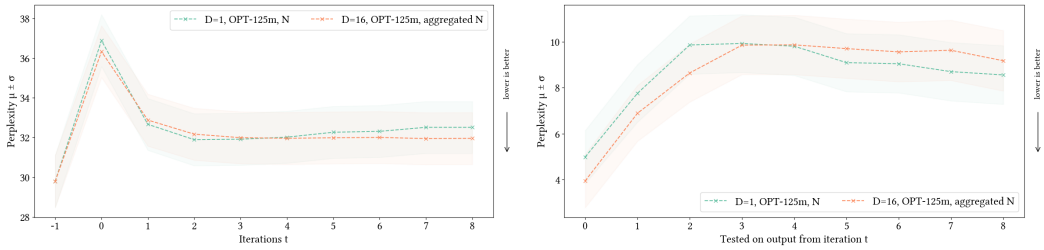


(a) **From the first iteration step ($t = 0$) onward, two community models consistently achieve lower perplexity than a single model trained and evaluated on the combined datasets.** $M = 1$: a single model trained and evaluated on the combination of Wikitext2 and Penn Treebank (green). $M = 2$: two community models, each trained and evaluated on its corresponding dataset (orange). To ensure a controlled comparison, the length of the Wikitext2 dataset is reduced to match the length of Penn Treebank.



(b) **Two models trained on two different segments of Wikitext2 consistently achieve lower perplexity than two models both trained on one of the two segments.** $D = 1$: Two models trained on identical segment 0 (green). $D = 2$: Two models trained on two different segments 0 and 1 (orange), corresponding to the same ecosystem as M=2 in the primary setting. Unlike in the primary segmentation approach, diversity is achieved through expansion of the distribution.

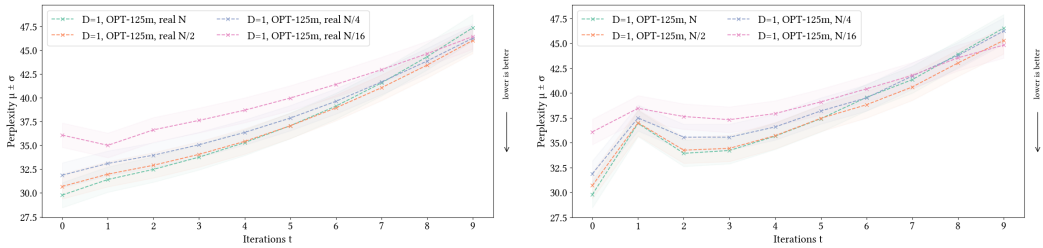
Fig. 17. The shaded region is a combined standard deviation across the models and data points.



(a) OPT-125m separately trained on generated output

(b) OPT-125m tested on generated output

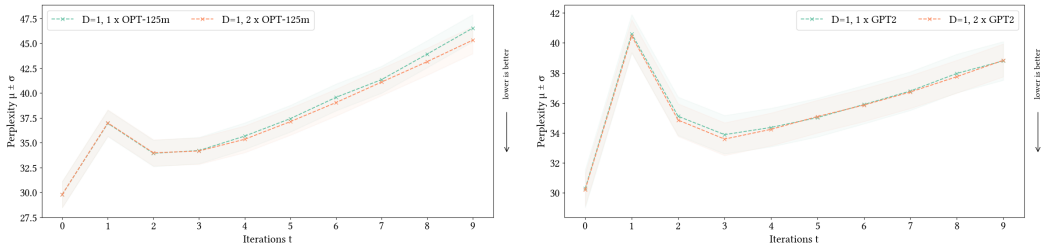
Fig. 18. **Data quality alone cannot explain performance differences across ecosystems.** The OPT-125m base model was separately trained (a) and tested (b) on generated outputs from the least and most diverse ecosystems ($D = 1$ and $D = 16$, respectively). For $D = 16$, outputs from all 16 models were aggregated. In (a), the off-the-shelf OPT-125m base model was fine-tuned separately on each generated dataset and subsequently evaluated on the fixed Wikitext2 test set. The similar perplexity values obtained for $D = 1$ and $D = 16$ indicate that model-independent data quality cannot fully account for the performance differences observed in our main experiments. Lower perplexity values for outputs from iterations $t = 0$ and $t = 1$ in (b) suggest that artificial data generated at early stages of the self-training loop more closely resembles the pretraining distribution of the base OPT-125m model. Fine-tuning on such artificial data shifts the distribution of the fine-tuned model back toward the pretraining distribution and away from the true distribution, which helps explain the performance drop observed with the output from iteration $t = 0$ in (a) and consistently across our experiments.



(a) Single OPT-125m re-trained on the real Wikitext2 train set

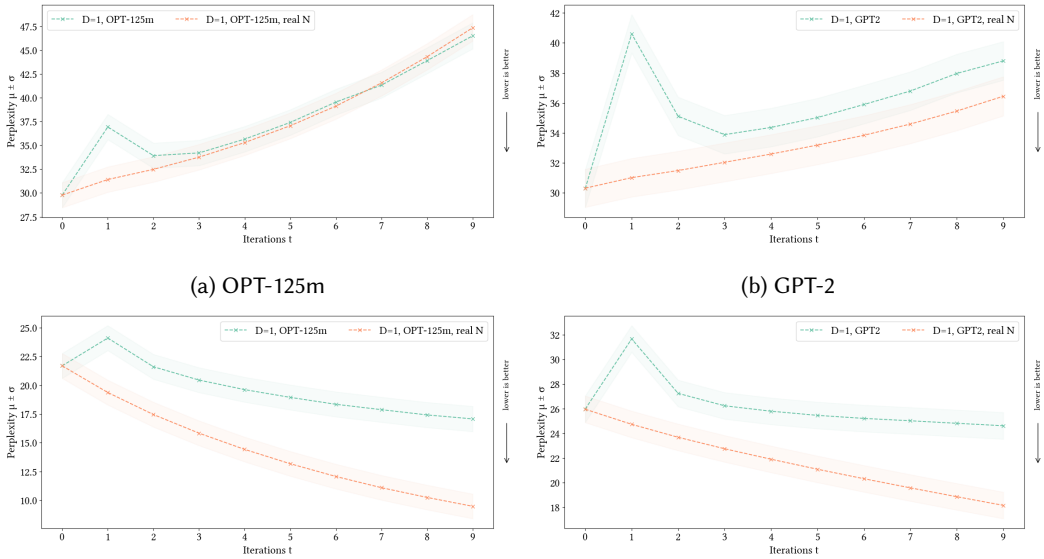
(b) Single OPT-125m self-trained on own output

Fig. 19. **Training data size alone cannot explain performance differences across ecosystems.** A single OPT-125m model was analyzed in two conditions: (a) re-trained on the real Wikitext2 training set and (b) self-trained on its own output, respectively. The initial training data content and sizes correspond to the primary setting ($N = 0.8$M tokens), but only one model was used in each run. The condition "$D = 1$, OPT-125m, N" matches $D = 1$ in the primary setting. The similar perplexity values observed across different data sizes indicate that training data size alone cannot account for the performance differences across ecosystems.



(a) OPT-125m (b) GPT-2

Fig. 20. **Number of models cannot fully explain the performance differences observed across ecosystems.** 1×OPT-125m and 1×GPT-2: a single model trained on the entire dataset of size N, corresponding to the $D = 1$ condition in the primary setting (green). 2×OPT-125m and 2×GPT-2: two identical models, each trained on the entire dataset of size N (orange). The generation datasets are identical for all models. The training datasets are identical at the first iteration step but may differ slightly in later iterations for the two-model runs due to stochasticity in setup. The shaded region represents the combined standard deviation across models and data points.



(a) OPT-125m (b) GPT-2

(c) OPT-125m evaluated on train set (d) GPT-2 evaluated on train set

Fig. 21. **Overfitting alone cannot explain the observed performance decay.** Comparison of the self-training loop in the primary setting (green) with a re-training loop on the fixed real training set (orange), evaluated on the test set (top) and training set (bottom). Because the training data in the re-training condition remains fixed, any performance decay can be fully attributed to overfitting. The lower perplexity on the training set in the re-training condition indicates stronger (over)fitting to the fixed training data. This suggests that overfitting alone cannot account for the performance decline observed in the self-training loop, which appears comparable for OPT-125m and even stronger for GPT-2.